

Vignette: SampleOptim

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1 Introduction

SampleOptim was developed and implemented as part of the Portuguese National Programme for Biological Sampling (EU Data Collection Framework). Its primary objective is to determine the optimal number of fish per length class to sample, ensuring accurate input data (e.g., ALKs – age-length keys and MO – maturity ogive) for species stock assessment. The key feature of SampleOptim is its ability to run simulations, enabling the consideration of various stratification conditions. These include temporal stratification (annual, semester, or quarterly), spatial stratification like port selection (uniform or random selection of ports or fishing gears), and sex stratification (the tool allows to define the sex ratio within the subsample). In species with pronounced sexual dimorphism, sex stratification is a critical consideration, as the composition of females and males within the subsample may significantly impact the accuracy of the Age-Length Key (ALK) derived from simulation outputs. The tool was started to be developed in 2013 at the Workshop on Sampling Design and Optimization (WKSDO) (Azevedo et al. 2014), was updated in 2017, at the Workshop on Sampling Effort for Biological Parameters (WKSEBP) (Silva et al. 2017). In 2018, was presented at WKBIOPTIM2 (ICES, 2019a) and tested in case studies presented in the following report Sections: 4.2, 5.4 and 6.1. This tool, since 2018, was continually being used in cases studies and being improved in applications at WKBIOPTIM2-4 and WGBIOPTIM (ICES, 2019b; ICES, 2021; ICES, 2024).

The simulation process operates at the sample level, relying on a dataset that represents the “whole” population. Simulations are conducted using random subsampling without replacement, although users have the option to sample with replacement. The ICES WKBIOPTIM4 report (Section 4.2) provides an example comparing the results of both subsampling methods (ICES. 2021).

Estimates for ALKs (age-length keys) and MOs (maturity ogives) are generated and analyzed based on a reduced number of individuals sampled per length class. However, if the original sample size is insufficient, the algorithm functions may fail to converge, potentially indicating bias in the initial data sampling.

A series of quality indicators are estimated to assess the different scenarios tested and support the decision-making process in determining the optimal sample size by length class, considering that ALKs (age-length keys) and MOs (maturity ogives) are essential for stock assessment.

The following variables are generated to compare the original and simulated datasets: mean length-at-age, mean age-at-length, mean weight-at-length, mean weight-at-age, coefficient of variation for length-at-age and age-at-length, standard deviation for length-at-age and age-at-length, parameters of the von Bertalanffy growth model (L_{inf} , t_0 and k), and maturity ogive parameters (L_{25} , L_{50} and L_{75}).

In addition to the previously mentioned quality indicators, the root mean squared prediction error (*RMSPE*), mean squared prediction error (MSPE or MSE), and mean absolute percentage error (*MAPE*) are also calculated (ICES, 2019a; ICES, 2019b; ICES, 2021; ICES, 2024).

SampleOptim is a valuable R-tool for determining the optimal number of individuals to sample per length class. It also allows for evaluating and testing the impact of changes in sampling stratification—based on time period, port, fleet, sex, age-length keys, weight-at-age and maturity ogives.

The simulation algorithm for sample selection is the R-code file named as: `sample_selection_function.R`. The code has been divided in 3 R-code parts: 1) Data exploratory analysis (`1_Data_exploratory_analysis.R`); 2) Simulation process, setup of simulation scenarios and produce the simulation results data objects (`2_Simulations_vers1.R`); 3) Simulations statistical analysis (`3_Simulations_results_data_analysis_vers1.R`).

SampleOptim works using the subsampling without replacement by default, but could also work by subsampling with replacement. An example of the application of those two type of subsampling, with or without replacement was present at WKBOPTIM4 in Section 4.2 (ICES. 2021).

SampleOptim was designed to run automatically, requiring minimal user intervention aside from verifying the format of input data. The simulation resets, tables and plots, are automatically saved in separated folders.

The flowchart with the schematic approach behind SampleOptim simulation algorithm and data statistical analysis is illustrated in Figure 1.

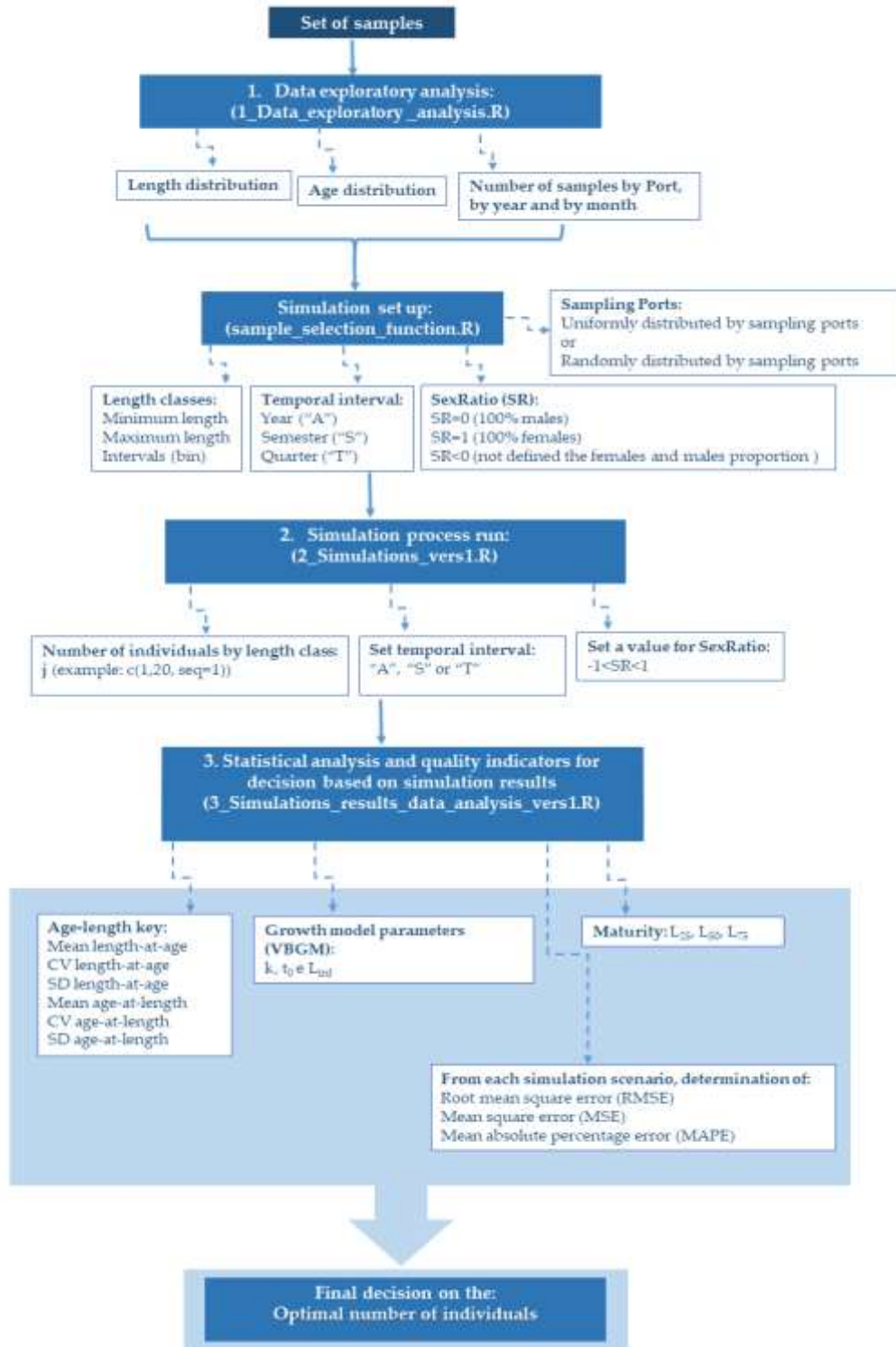


Figure 1 - Flowchart of the SampleOptim R-scripts with the main specifications, inputs, statistical analysis and quality indicators produced for applying on the decision process on the optimal number of individuals to sample by length class.

2 Data

The R-tool uses a dataset in .csv file format as input data, requiring the following mandatory columns:

```
##
## Attaching package: 'data.table'

## The following object is masked from 'package:raster':
##
##      shift

## The following objects are masked from 'package:dplyr':
##
##      between, first, last

## The following object is masked from 'package:reshape':
##
##      melt

## [1] "ID_BIO_FISH"      "date"              "Month"              "Year"
## [5] "Species"          "Port"              "LengthClass"        "Weight"
## [9] "Sex"              "MaturityStage"     "Age"
```

Description of the variables on the input dataset for SampleOptim simulations R-tool:

ID_BIO_FISH: Integer (unique and different numeric value for each line)

date: Factor

month: Integer (month of sampling; numeric value from 1 to 12)

year: Integer (year of sampling, four numbers format)

COD_FAO: Integer (three letter FAO-code for the species)

Port: Factor (Port/harbour name; or Region/Area of sampling)

Length_class: Integer (length class measurement, units: mm, cm, m)

Weight: Numeric (units: Kg, g)

Sex: Factor (F - females; M - males; I - undetermined sex)

Maturity_stage: Integer or character (according to the species maturity stage key)

Age: Integer

Each row in this dataset represents a single individual. SampleOptim provides a function to convert the old Regional Database CS-CA tables (RDB exchange format) into the required input format. This conversion function can be found in the R script titled: "A.Data conversion and preparation.R".

The new Regional Database and Estimation System (RDBES) data model provides detailed sampling information, including the sampling designs implemented by each EU Member State. These designs are organized into what are known as sampling hierarchies. The corresponding exchange format and documentation are described in the public repository of the RDBES development [GitHub](#). The required data can be downloaded directly from the homepage [RDBES](#).

RDBES officially replaced the previous Regional Database [RDB](#) data system in 2023. However, historical data continue to be stored in the RDB, which remains — for now — the default format used by all WGBIOPTIM Rtools.

3 Indicators

To enable comparison across the different simulation sets generated for each scenario, a set of quality indicators is estimated. These indicators help evaluate the alternatives and support the decision-making process for identifying the optimum sample size by length class.

Since these data will serve as input for assessments—specifically for the ALKs and MO parameters—the following estimates are compared between the original dataset and the simulated data subsets: mean length-at-age, mean age-at-length, the von Bertalanffy growth model parameters (L_{inf} , t_0 and k) and the maturity ogive parameters (L_{25} , L_{50} and L_{75}).

In addition to these indicators, the root mean squared prediction error ($RMSPE$), mean squared prediction error ($MSPE$), and mean absolute percentage error ($MAPE$) are also calculated.

3.1 von Bertalanffy growth model (VBGM)

It is the most widely used growth model and is especially important in fisheries studies, von Bertalanffy derived this equation in 1938 his growth equation in 1938 based on physiological principles that explain how organisms grow over time, the approach was grounded in biological reasoning rather than purely mathematical modeling. The model assumes that the rate of growth of an organism declines with size so that the rate of change in length, l , may be described by:

$$\frac{dl}{dt} = k(L_{inf} - l)$$

where: t is time; l is length (or some other measure of size); k is growth rate, describing the rate at which the organism approaches its maximum length; L_{inf} is the theoretical maximum length the organism can reach (asymptotic length) at which growth is zero. By integrating the previous equation, it becomes:

$$lt = L_{inf}(1 - e^{-k(t-t_0)})$$

The parameter $t0$ is defined as the time at which the organisms would have had zero size. Thus to fit this equation by nonlinear regression, is necessary to provide approximate guesses as starting points for the 3 parameters ($Linf$, k and $t0$) . These help guide the fitting process.

3.2 Maturity ogive (MO)

A generalized linear model (GLM) with a binomial error distribution and a logit link was fitted to the proportion of fish mature by length class. The model has the general form:

$$\log(\mathbb{E}[m]) = a + bl$$

where a and b are the intercept and slope of the ogive. Estimates of the length at which 50% of the individuals were mature ($L50$) and of the slope of the ogive at $L50$ were derived from the model parameters as:

$$L50 = \frac{-b}{a}$$

A similar procedure as applied to estimate $L25$ and $L75$.

3.3 Mean squared prediction error (MSPE)

$MSPE$ is a statistical measure that quantifies the average squared difference between predicted and actual values, providing insight into the accuracy of a predictive model. It's calculated by summing the squared differences between each predicted value and its corresponding actual value, then dividing by the number of observations. The $MSPE$ is the mean square error of the estimator, if $\hat{\theta}$ is an estimator of the parameter θ , then

$$MSPE(\theta) = \mathbb{E}[\hat{\theta} - \theta]$$

3.4 Root mean squared prediction error (RMSPE)

$RMSPE$ is a statistical measure used to evaluate the accuracy of predictions made by a model. It provides a single value summarizing how far the predicted values deviate from the actual (observed) values, emphasizing larger errors due to the squaring process. The $RMSPE$ is the square-root of the mean squared error ($MSPE$) of the estimator. In symbols, if $\hat{\theta}$ is an estimator of the parameter θ , then

$$RMSPE(\theta) = \sqrt{\mathbb{E}[\hat{\theta} - \theta]}$$

The $RMSPE$ of an estimator is a measure of the expected error of the estimator. The units of $RMSPE$ are the same as the units of the estimator.

3.5 Mean absolute percentage error (MAPE)

The mean absolute percentage error (*MAPE*), also known as mean absolute percentage deviation (*MAPD*), measures the accuracy of a method for constructing fitted time series values in statistics. It calculates the mean absolute percentage error (Deviation) function for the forecast and the eventual outcomes. The mean absolute percentage error (*MAPE*) is defined as:

$$\text{MAPE} = 100 \times \frac{1}{N} \sum_{i=1}^N \left| \frac{x_i - \hat{x}_i}{x_i} \right|$$

where x_i is the actual observations time series, $\hat{x}_i$ is the estimated or forecasted time series and N is the number of non-missing data points.

4 Worked examples

Steps of a practical performance of the proposed tool are provided in the example below.

4.1 Data (R-script: 1_Data_exploratory_analysis.R)

```
source("SampleOptim/1_Data_exploratory_analysis.R")
```

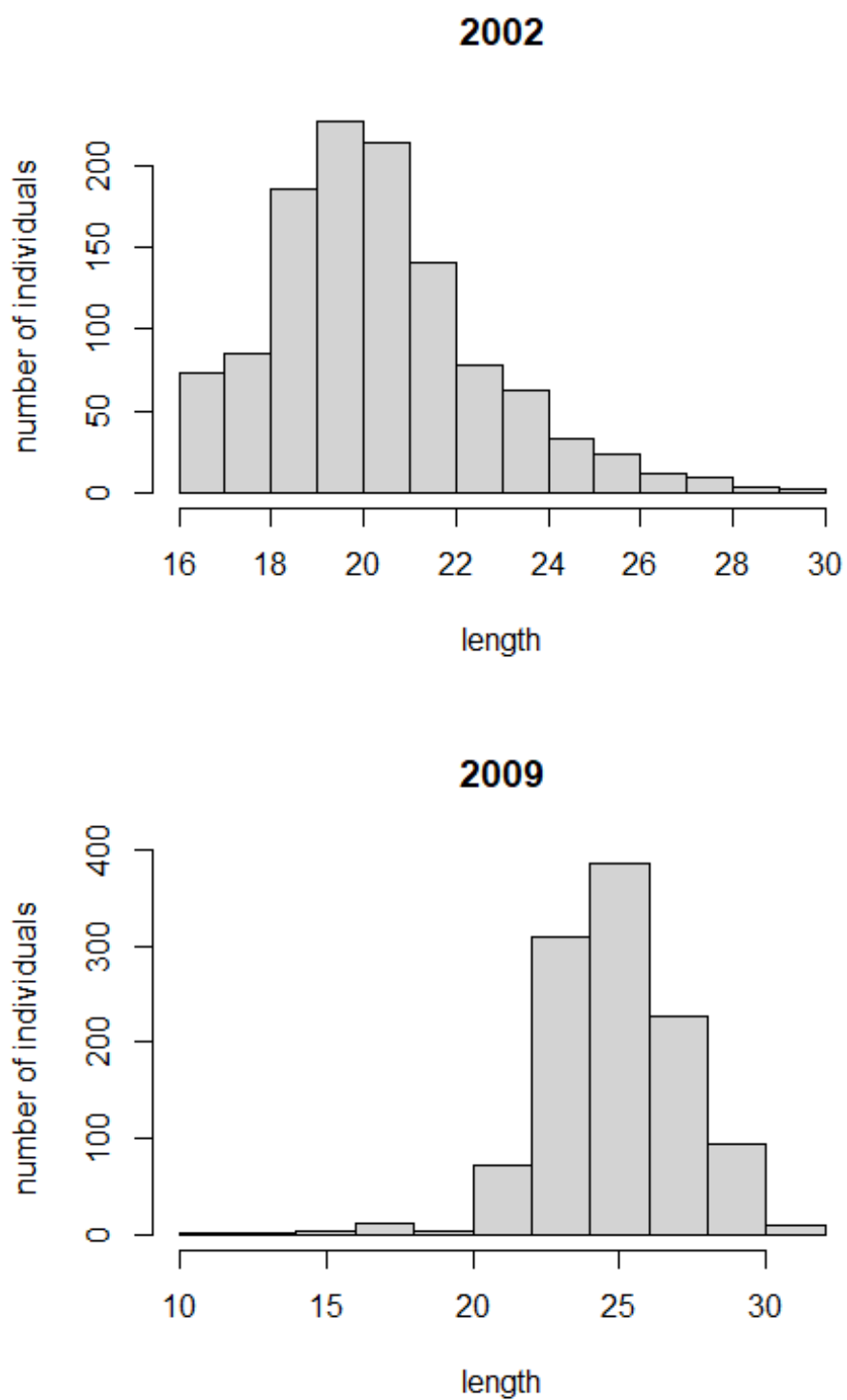


Figure 2 - Length distribution of blue whiting samples for 2002 and 2009.

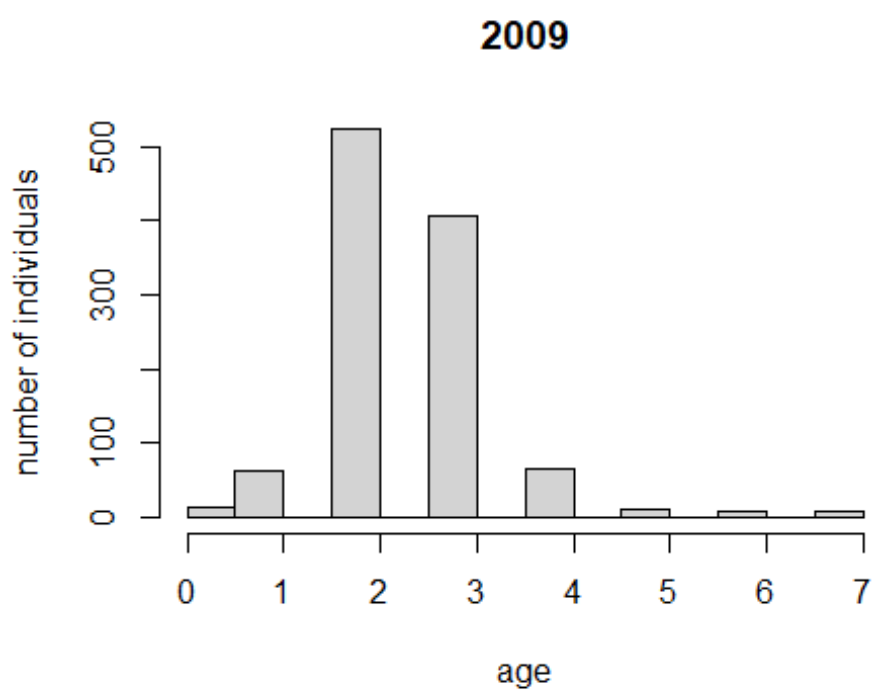
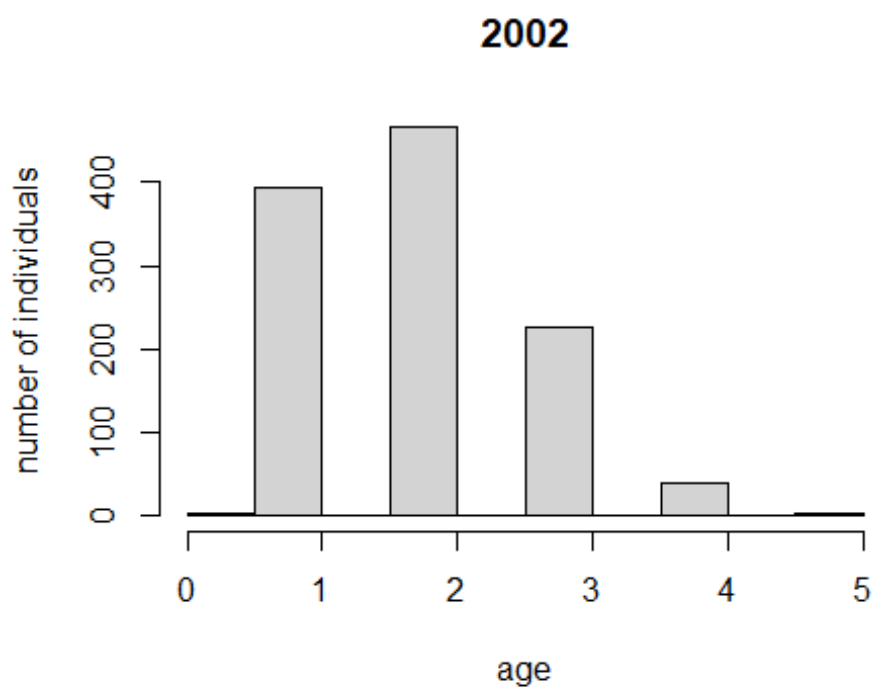


Figure 3 - Age distribution of blue whiting samples for 2002 and 2009.

```

data<-read.table("SampleOptim_input_fileformat_example.csv",sep=";",
header=T)
#Confirm sep = "," or = ";"

# Show summary
summary(data)

##   ID_BIO_FISH      date      Month      Year
##   Min.      :257949   Length:2263   Min.      : 1.000   Min.      :2002
##   1st Qu.:270349   Class :character   1st Qu.: 3.000   1st Qu.:2002
##   Median :274627   Mode  :character   Median : 6.000   Median :2002
##   Mean    :404383           Mean    : 6.011   Mean     :2005
##   3rd Qu.:542379           3rd Qu.: 9.000   3rd Qu.:2009
##   Max.     :551969           Max.     :12.000   Max.     :2009
##
##   Species      Port      LengthClass      Weight
##   Length:2263   Length:2263   Min.      :11.00   Min.      : 8.06
##   Class :character   Class :character   1st Qu.:20.00   1st Qu.: 51.16
##   Mode  :character   Mode  :character   Median :23.00   Median : 76.36
##                               Mean    :23.02   Mean    : 82.57
##                               3rd Qu.:25.00   3rd Qu.:107.13
##                               Max.     :32.00   Max.     :266.92
##                               NA's      :2
##
##   Sex      MaturityStage      Age
##   Length:2263   Length:2263   Min.      :0.000
##   Class :character   Class :character   1st Qu.:2.000
##   Mode  :character   Mode  :character   Median :2.000
##                               Mean    :2.211
##                               3rd Qu.:3.000
##                               Max.     :7.000
##                               NA's      :33

```

4.2 Simulations (R-script: 2_Simulations.R)

Simulation Input Condition Parameters

Variable. name	Mandatory	Variable. values	Definition
species	y	WHB	CODE_FAO
AREA	y	27.9.a	

Variable. name	Mandatory	Variable. values	Definition
AGE_ONL Y	y	FALSE	TRUE- only statistical analysis for age; FALSE - includes age and maturity data analysis
PORT	n	FALSE	Uses Port stratification for subsampling (TRUE); do not take into account the Port Stratification (FALSE)
distUniPo rto	n	FALSE	Uniform distribution of subsamples by Port (TRUE); random distribution of subsamples by Port (FALSE)
TIME_ST RATA	y	T	A - year; S- semester; T - quarter
SEX_RATI O	y	0.5	0 - only males; 1 - only females
MIN_LC	y	13	minimum length class
MAX_LC	y	36	maximum length class
interval_L C	y	1	length class step
MIN_age	y	0	minimum age
MAX_age	y	7	maximum age
MIN_OTOL	y	1	minimum number of individuals by length class in the simulation setup
MAX_OTOL	y	10	maximum number of individuals by length class in the simulation setup
interval_ OTOL	y	1	interval number of individuals by length class in the simulation setup
EXTRA_OTOL	n	15 20	extra otoliths that are not within MIN_OTOL and MAX_OTOL
Linf	y	45	Von Bertalanffy growth model parameter - Linf. Used as a starting value to adjust VBGM.
K	y	0.1	Von Bertalanffy growth model parameter - k. Used as a starting value to adjust VBGM.
t0	y	-3	Von Bertalanffy growth model parameter - t0. Used as a starting value to adjust VBGM.
year_start	y	2009	first year data subset to run simulations
year_end	y	2009	last year data subset to run simulations
stage_mat ure	y	2	define the maturity stages that correspond to mature stages (to allow to determine the proportion of immatures and matures)

Variable. name	Mandatory	Variable. values	Definition
n	y	100	define the number of simulations (bootstrap runs)

4.3 Results (R-script: 3_Simulations_results_data_analysis.R)

4.3.1 Mean length-at-age (ALK comparison)

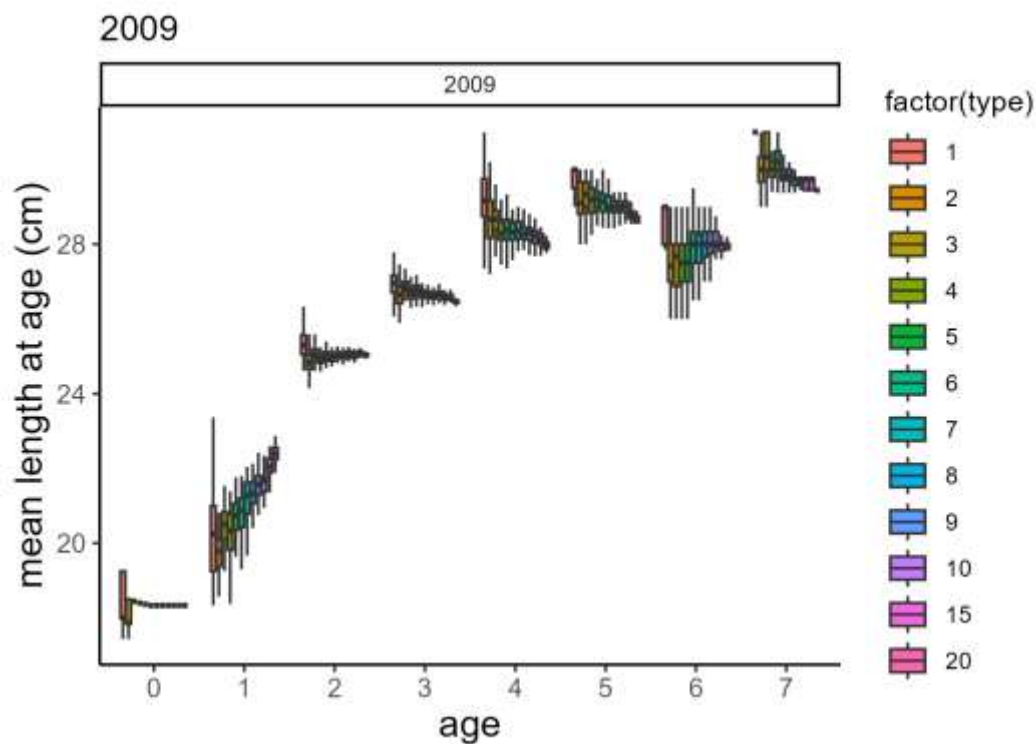


Figure 4 - Mean length (cm) at age of all the iterations (n=100) from the simulation scenarios applied to the blue whiting sampling from 2009, for the following number of individuals selected by length class (factor(type) is the number of individuals by length class defined as $j=1,2,3,4,5,6,7,8,9,10,15,20$).

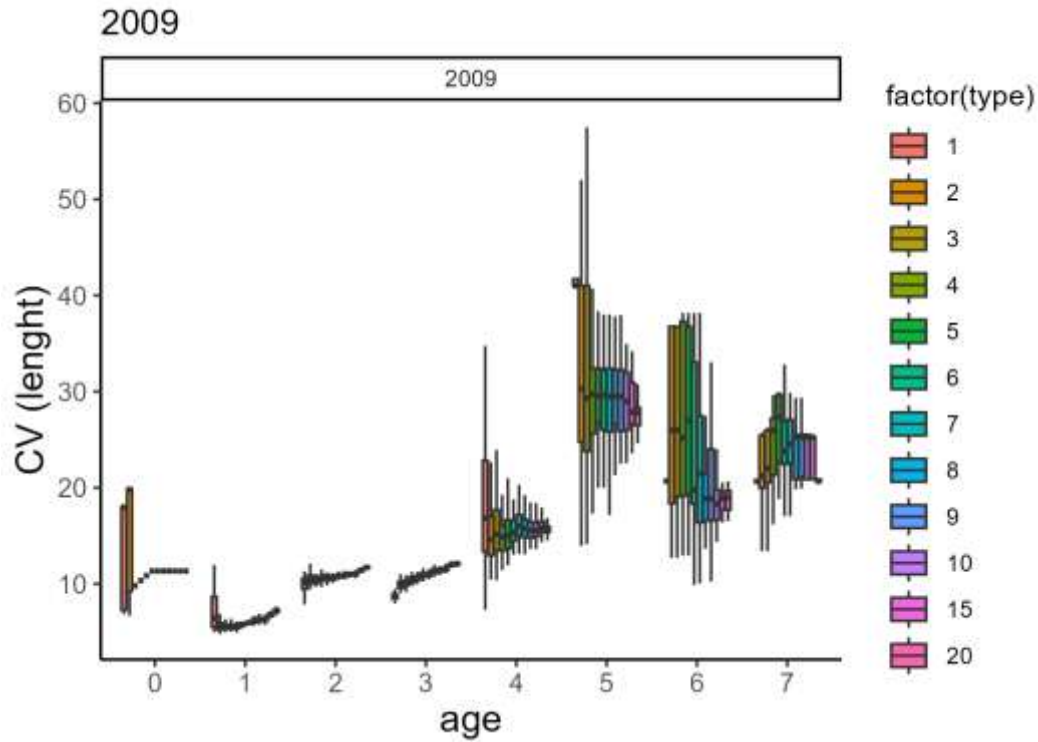


Figure 5 - Coefficient of variation (CV) of the length (cm) at age of all the iterations ($n=100$) from the simulation scenarios applied to the blue whiting sampling from 2009, for the following number of individuals selected by length class (factor(type) is the number of individuals by length class defined as $j=1,2,3,4,5,6,7,8,9,10,15,20$).

4.3.2 Mean weight-at-age

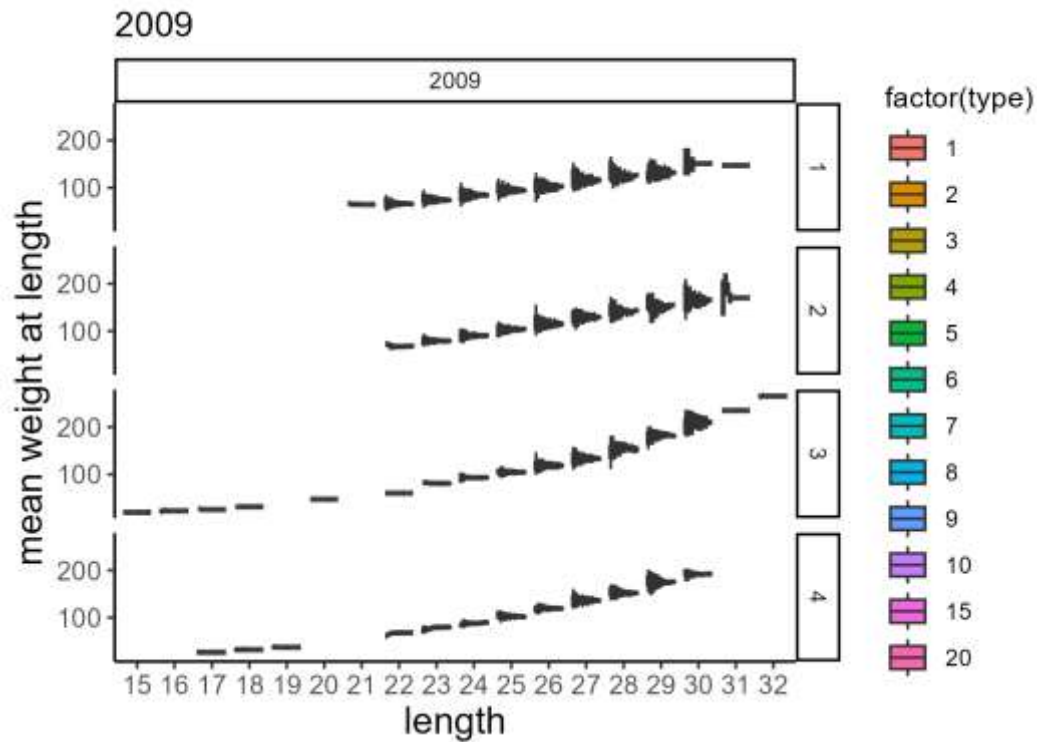


Figure 6 - Mean weight (g) at length of all the iterations (n=100) from the simulation scenarios applied to the blue whiting sampling from 2009, for the following number of individuals selected by length class (factor(type) is the number of individuals by length class defined as $j=1,2,3,4,5,6,7,8,9,10,15,20$).

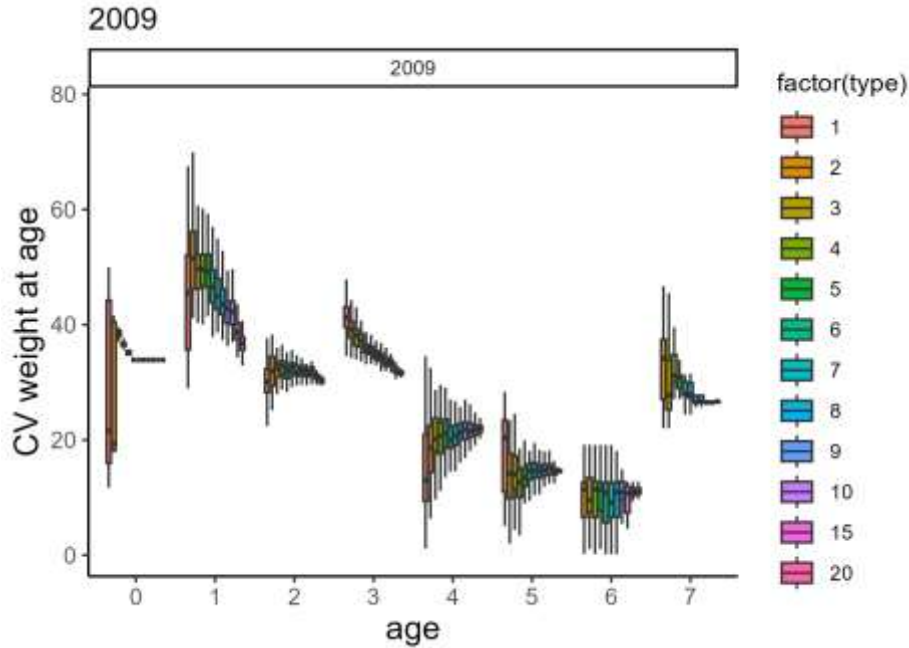


Figure 7 - Coefficient of variation (CV) of weight (g) by age group of all the iterations (n=100) from the simulation scenarios applied to the blue whiting sampling from 2009, for the following number of individuals selected by length class (factor(type) is the number of individuals by length class defined as $j=1,2,3,4,5,6,7,8,9,10,15,20$).

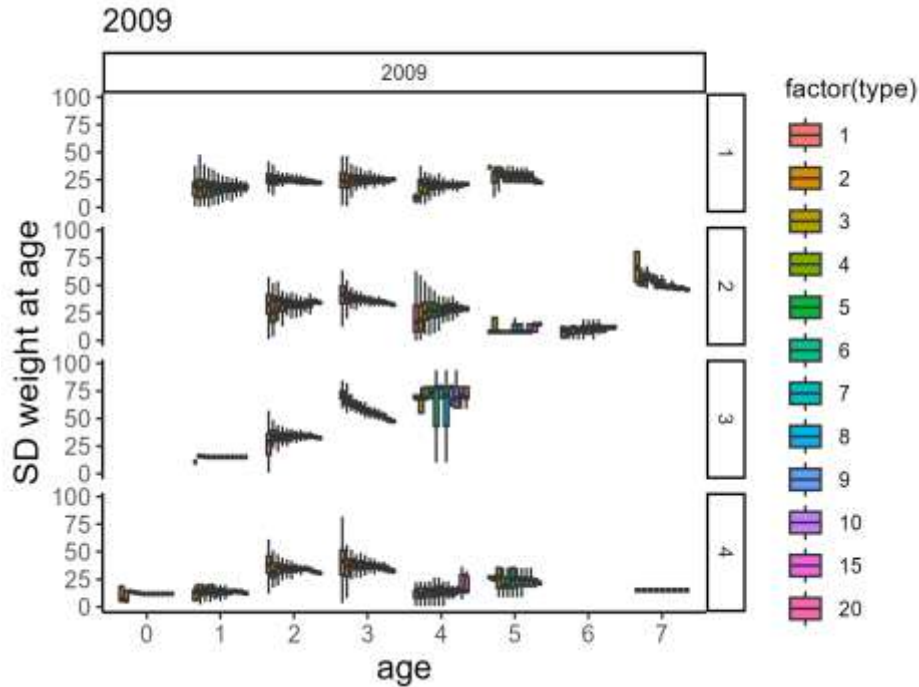


Figure 8 - Standard desviation of weight (g) at age of all the iterations (n=100) from the simulation scenarios applied to the blue whiting sampling from 2009, for the following number of individuals selected by length class (factor(type) is the number of individuals by length class defined as j=1,2,3,4,5,6,7,8,9,10,15,20).

4.3.3 von Bertalanffy growth model results (*Linf*, *t0* and *k* comparison)

```
knitr::include_graphics("SampleOptim\\output\\Fig7_t0_VBGM_T.png")
```

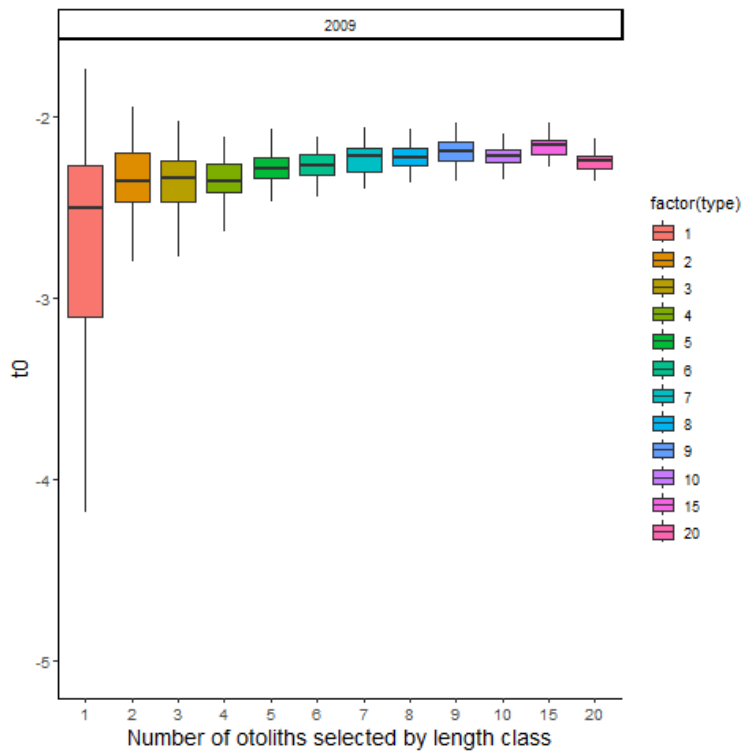


Figure 9 - Time when length as zero (t_0) of all the iterations ($n=100$) from the simulation scenarios applied to the blue whiting sampling from 2009, for the following number of individuals selected by length class (factor(type) is the number of individuals by length class defined as $j=1,2,3,4,5,6,7,8,9,10,15,20$).

```
knitr::include_graphics("SampleOptim\\output\\Fig7_K_VBGM_T.png")
```

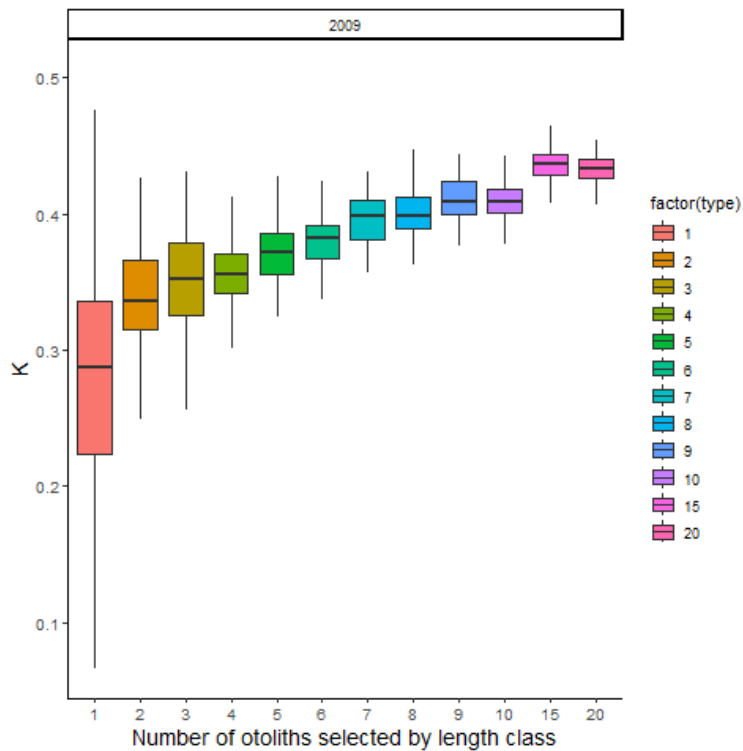


Figure 10 - Growth rate coefficient (k) of all the iterations ($n=100$) from the simulation scenarios applied to the blue whiting sampling from 2009, for the following number of individuals selected by length class (factor(type) is the number of individuals by length class defined as $j=1,2,3,4,5,6,7,8,9,10,15,20$).

```
knitr::include_graphics("SampleOptim\\output\\Fig7_Linf_VBGM_T.png")
```

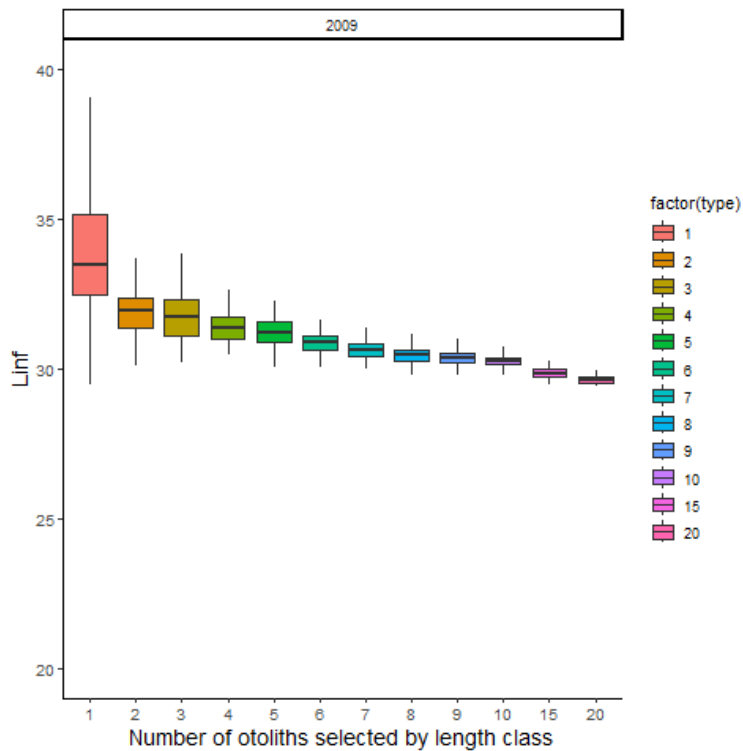


Figure 11 - Asymptotic length (L_{inf}) of all the iterations ($n=100$) from the simulation scenarios applied to the blue whiting sampling from 2009, for the following number of individuals selected by length class (factor(type) is the number of individuals by length class defined as $j=1,2,3,4,5,6,7,8,9,10,15,20$).

4.3.4 Maturity ogive (L_{25} , L_{50} and L_{75} comparison)

```
knitr::include_graphics("SampleOptim\\output\\simulation_results\\FigmaLt_2009.png")
```

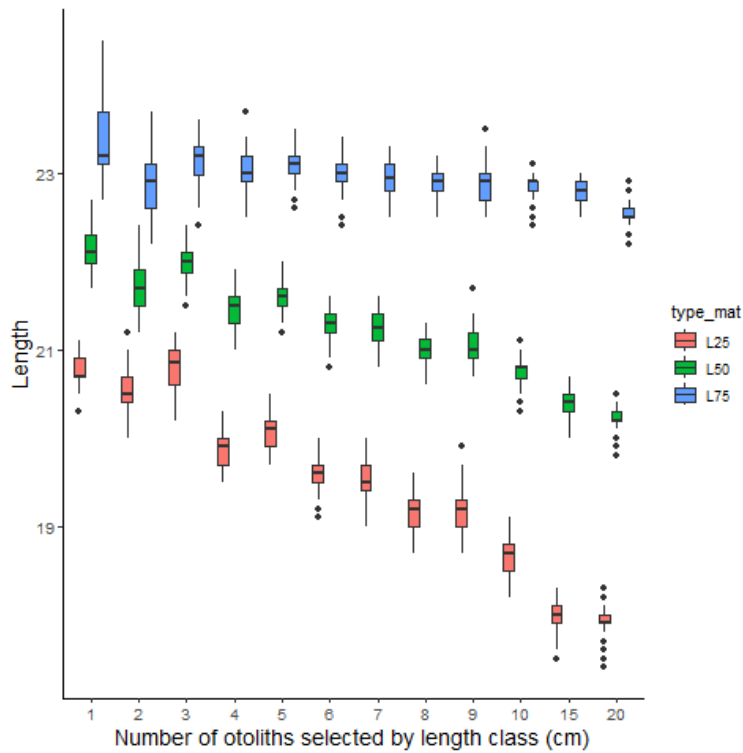


Figure 12 - The length at each 25%, 50% and 75% (type_mat=L25, L50, L75) of the individuals were mature, resulting from the iterations (n=100) from the simulation scenarios applied to the blue whiting sampling from 2009, for the following number of individuals selected by length class (factor(type) is the number of individuals by length class defined as j=1,2,3,4,5,6,7,8,9,10,15,20).

4.3.5 Mean squared prediction error (MSPE)

```
knitr::include_graphics("SampleOptim\\output\\Fig10_norm_mn_mspe_T.png")
```

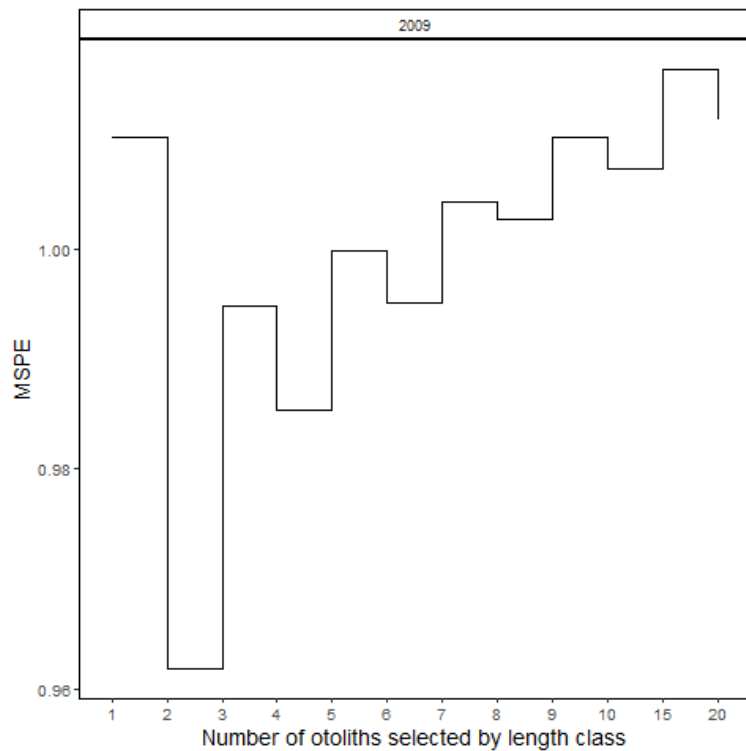


Figure 13 - Mean square prediction error (MSPE) resulting from the iterations (n=100) from the simulation scenarios applied to the blue whiting sampling from 2009, for the following number of individuals selected by length class (j=1,2,3,4,5,6,7,8,9,10,15,20).

4.3.6 Root mean squared prediction error (RMSPE)

```
knitr::include_graphics("SampleOptim\\output\\Fig10_norm_mn_rtmspe_T.png")
```

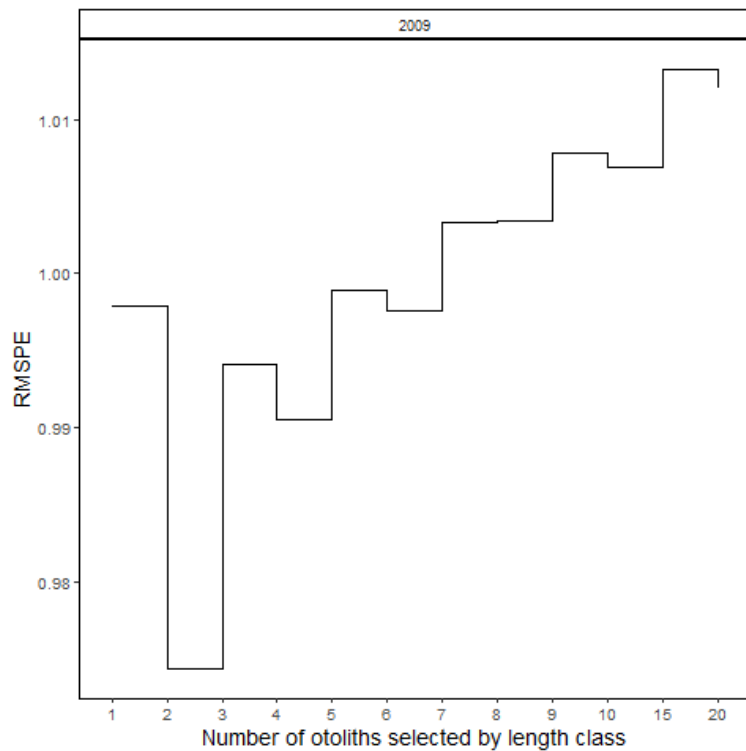


Figure 14 - Root mean squared prediction error (RMSPE) resulting from the iterations ($n=100$) from the simulation scenarios applied to the blue whiting sampling from 2009, for the following number of individuals selected by length class ($j=1,2,3,4,5,6,7,8,9,10,15,20$).

4.3.7 Mean absolute percentage error (MAPE)

```
knitr::include_graphics("SampleOptim\\output\\Fig10_norm_mn_mape_T.png")
```

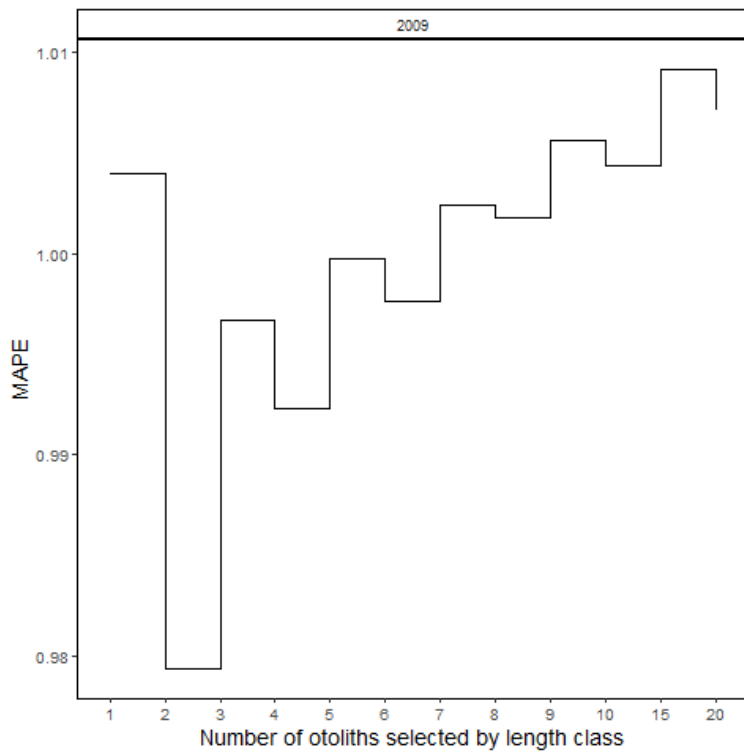


Figure 15 - Mean absolute percentage error (MAPE) resulting from the iterations (n=100) from the simulation scenarios applied to the blue whiting sampling from 2009, for the following number of individuals selected by length class (j=1,2,3,4,5,6,7,8,9,10,15,20).

The plots and the data tables with the estimated indicators parameters from the simulations are saved (Folders: output; and simulation_results), this allows the user to perform more analysis and if needed to include the results to choose the best optimization option in a separate table of the report. The main aim of the SampleOptim R-code scripts were to perform the simulation with different levels of stratification and have all the information needed to support end-users on their final decision considering the optimal sample size. SampleOptim is available at WBIOPTIM GitHub page under the "SampleOptim" folder.

In 2025, an updated modified version of SampleOptim was prepared. The updated test version of the toolbox was created and is available at WBIOPTIM GitHub page under the "SampleOptim_2025_3.0" folder..

Important note: the tests using the same input data set with the same simulations initial conditions and the comparison between the results obtained with original R tool SampleOptim and the 2025 modified version (SampleOptim_2025_3.0) *were not yet performed*. Simulation tests are planned to be presented at the 2026 WGBIOPTIM meeting.

5 References

Azevedo, M., Silva, C., Vølstad, J.H., Prista, N., Alpoim, R., Moura, T., Figueiredo, I., Dias, M., Fernandes, A.C., Lino, P., Felício, M., Chaves, C., Soares, E., Soares, S., Gonçalves, P., Costa, A.M., Nunes, C. 2014. Report of the Workshop on sampling design and optimization of fisheries data (WKSDO). Relat. Cient. Téc. IPMA, nº 2.

<https://www.ipma.pt/export/sites/ipma/bin/docs/publicacoes/ipma/rct-ipma-sd-2.pdf>

ICES. 2019a. Report of the Workshop on Optimization of Biological Sampling (WKBIOP- TIM 2). WKBIOPTIM 2 Report 2018 29–31 May 2018. Nantes, France. ICES CM 2018/EOSG:23. 172 pp

ICES. 2019b. Workshop on Optimization of Biological Sampling (WKBIOPTIM 3). ICES Scientific Reports. 1:78. 219 pp. <http://doi.org/10.17895/ices.pub.5647>

ICES. 2022. Fourth Workshop on Optimization of Biological Sampling (WKBIOPTIM4; Outputs from 2021 meeting). ICES Scientific Reports. 4:69. 80 pp. <http://doi.org/10.17895/ices.pub.21103000>

ICES. 2024. Working Group on Biological Sampling Optimization (WGBIOPTIM). ICES Scientific Reports. 6:51. 86 pp. <https://doi.org/10.17895/ices.pub.25941340>

6 GitHub links

https://github.com/ices-eg/wg_WGBIOPTIM/tree/BIOPTIM-R-tools/SampleOptim

<https://github.com/gonpatricia/SampleOptimRDBformat>

TESTED version (tests still needed): https://github.com/ices-eg/wg_WGBIOPTIM/tree/BIOPTIM-R-tools/SampleOptim_2025_3.0

7 Reproducibility information

```
## [1] "R version 4.3.3 (2024-02-29 ucrt)"

## R version 4.3.3 (2024-02-29 ucrt)
## Platform: x86_64-w64-mingw32/x64 (64-bit)
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##
##
## locale:
## [1] LC_COLLATE=Portuguese_Portugal.utf8
LC_CTYPE=Portuguese_Portugal.utf8
## [3] LC_MONETARY=Portuguese_Portugal.utf8 LC_NUMERIC=C
## [5] LC_TIME=Portuguese_Portugal.utf8
##
```

```

## time zone: Europe/Lisbon
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices utils      datasets  methods   base
##
## other attached packages:
## [1] data.table_1.17.4 raster_3.6-26      sp_2.1-3           tidyr_1.3.1
## [5] RCurl_1.98-1.14  boot_1.3-29       psyphy_0.3         MASS_7.3-
60.0.1
## [9] dplyr_1.1.2      cvTools_0.3.3     robustbase_0.99-2
lattice_0.22-5
## [13] ggthemes_5.1.0   ggplot2_3.5.2     reshape_0.8.9
nlstools_2.0-0
## [17] FSAdata_0.4.0    FSA_0.9.4         knitr_1.50
jsonlite_2.0.0
## [21] rmarkdown_2.29
##
## loaded via a namespace (and not attached):
## [1] generics_0.1.3   bitops_1.0-7      stringi_1.8.7
digest_0.6.37
## [5] magrittr_2.0.3   evaluate_1.0.3    grid_4.3.3
RColorBrewer_1.1-3
## [9] fastmap_1.2.0    plyr_1.8.8        purrr_1.0.4
scales_1.4.0
## [13] codetools_0.2-19 cli_3.6.1          crayon_1.5.2
rlang_1.1.6
## [17] withr_3.0.2      yaml_2.3.10       tools_4.3.3
vctrs_0.6.5
## [21] R6_2.6.1         png_0.1-8         lifecycle_1.0.4
stringr_1.5.1
## [25] pkgconfig_2.0.3  terra_1.7-71      pillar_1.10.2
gtable_0.3.6
## [29] glue_1.8.0       Rcpp_1.0.14       DEoptimR_1.1-3
xfun_0.52
## [33] tibble_3.2.1     tidyselect_1.2.0  rstudioapi_0.17.1
farver_2.1.2
## [37] htmltools_0.5.8.1 labeling_0.4.3    compiler_4.3.3

```